COOLANT CONTROL VALVE [SKYACTIV-D 1.5]

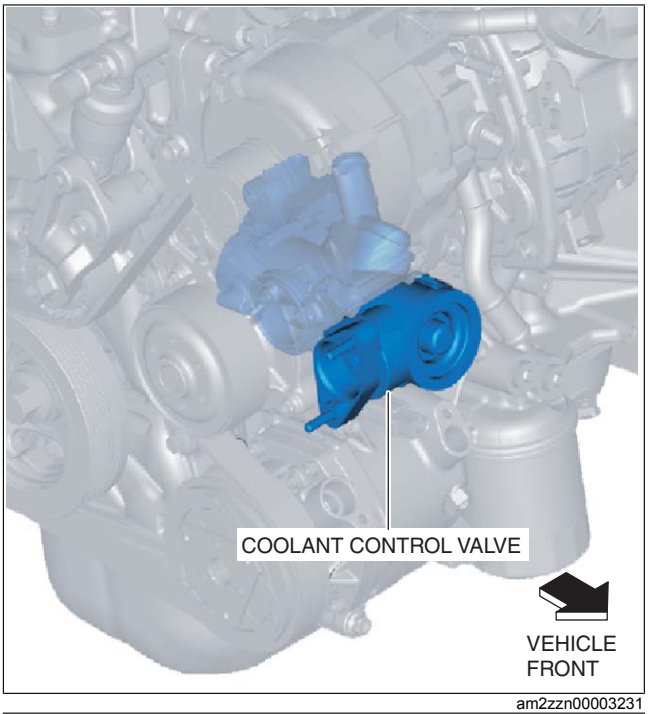
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Outline

- The engine coolant control valve supplies the necessary amount of engine coolant to the required passages, when needed, by adjusting the valve opening angle according to each of the engine operation conditions.
- By reducing cooling loss around the combustion chamber when the engine coolant temperature is low and ensuring combustion stability, fuel economy and emission performance have been improved.
- By steadily raising the engine coolant temperature, the temperature around the structure of the combustion chamber improves combustion, which has improved fuel economy and emission performance.
- By reducing the driving resistance of the water pump, mechanical resistance has been reduced and fuel economy improved.
- If the engine coolant temperature rises excessively due to an engine coolant control valve malfunction, the internal thermostat opens to allow engine coolant to flow to the cylinder heads, EGR cooler, LP-EGR control valve and the heater core to prevent overheating.

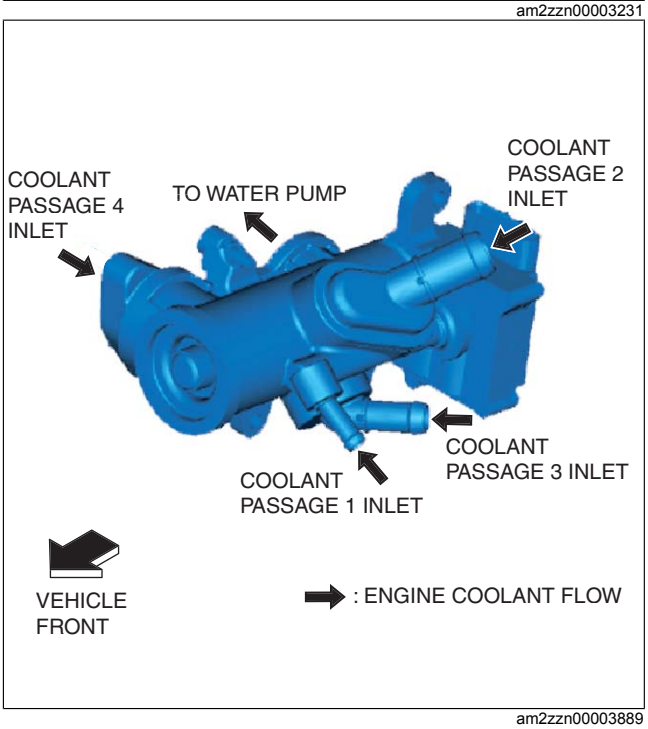
Construction

- The engine coolant control valve is located next to the water pump on the left, front side of the engine.

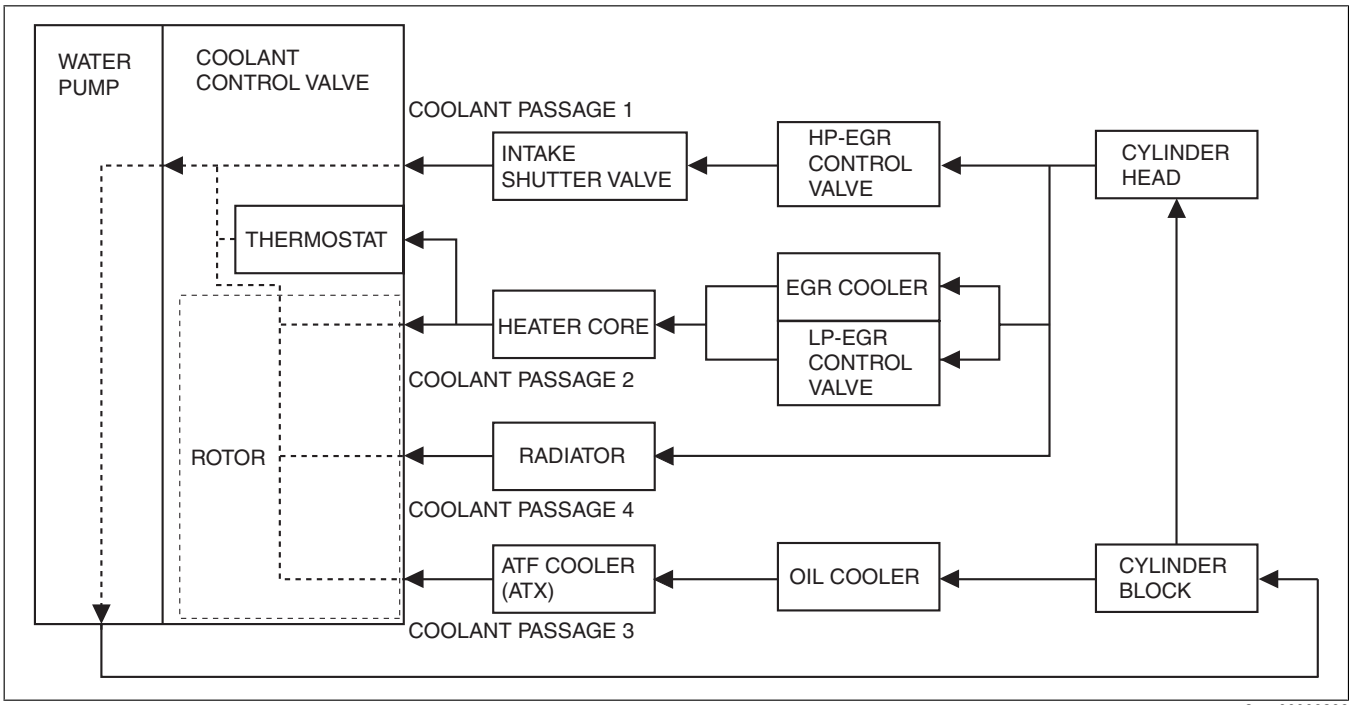


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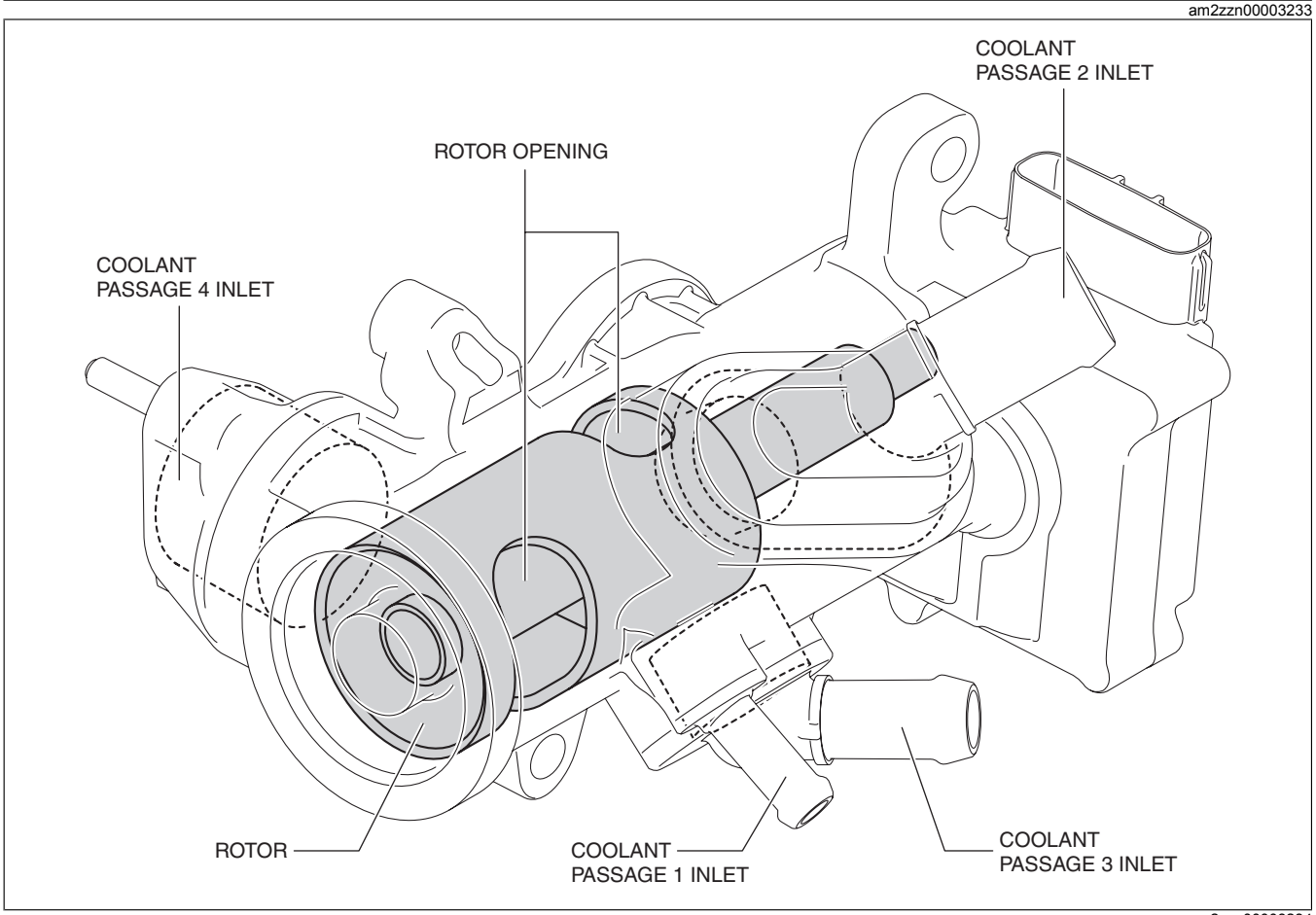
- Engine coolant flows from 4 engine coolant passages into the engine coolant control valve. There is one outlet to the water pump.



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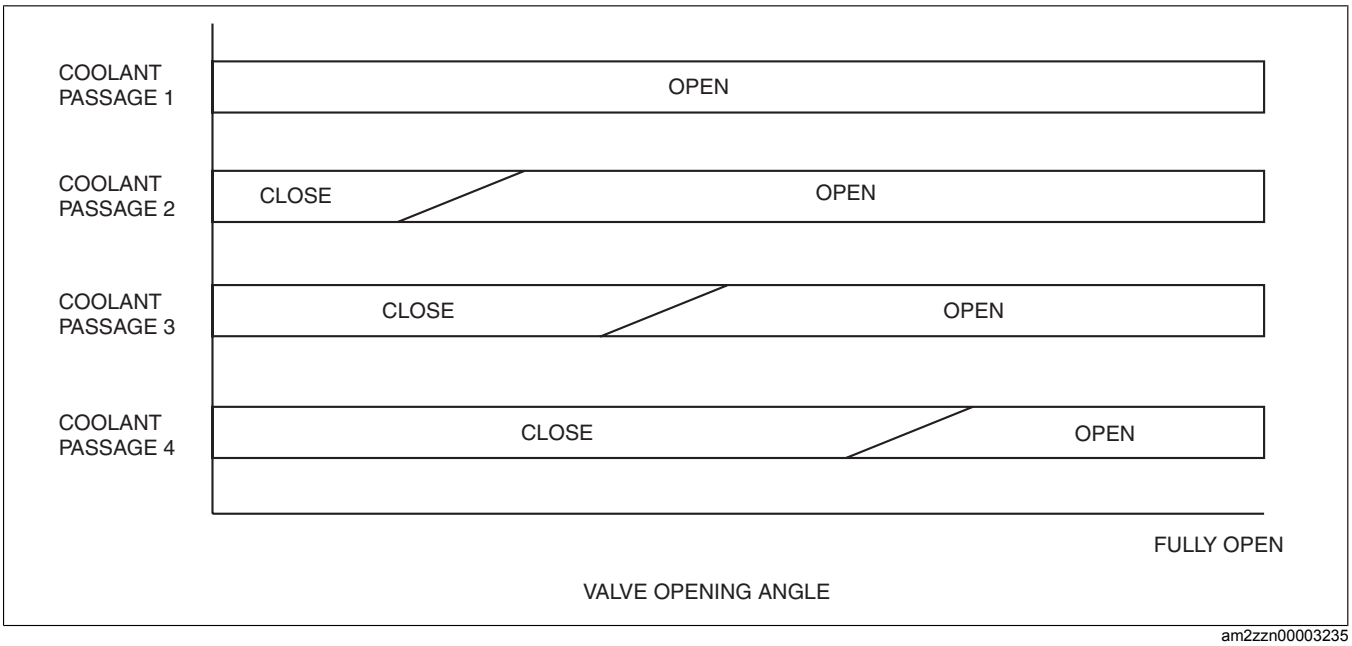


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- Coolant flows constantly in coolant passage 1.
- At coolant passages 2, 3, and 4, coolant flows when the rotor rotates and the rotor opening is aligned to the coolant passage inlet.



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Fail-safe

DTC No.	Fail-safe function
P0116:00	• PCM restricts engine torque. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • PCM restricts engine-transaxle integration control. (ATX) • Increase the idle speed. • Inhibits the A/C control. • Inhibits engine-stop by operating the i-stop function.
P2681:00	• PCM restricts engine torque. • Inhibits the EGR control. • Inhibits the auto diesel particulate filter regeneration control. • Inhibits the compulsory diesel particulate filter regeneration control. • Inhibits engine-stop by operating the i-stop function.
P26A3:00	